SAAT / GAT QUANTITATIVE (ENGLISH) • UNIT 2.01

C GAT Ratios, Percentages, Work & Motion Problems

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

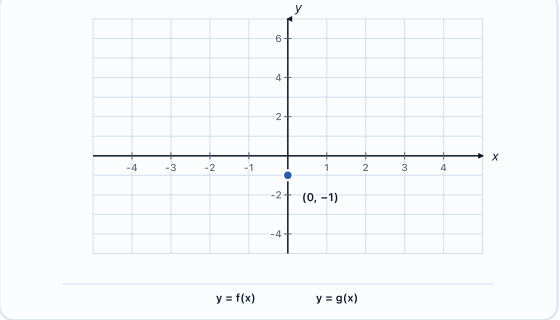
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 B	Q2 C	Q3 D	Q4 A	Q5 B
Q6 C	Q7 D	Q8 A	Q9 D	Q10 A
Q11 B	Q12 B	Q13 A	Q14 A	Q15 A
Q16 B	Q17 C	Q18 D	Q19 A	Q20 C
Q21 C	Q22 B	Q23 D	Q24 A	Q25 C

GAT Ratios, Percentages, Work & Motion Problems Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 8x + 28$ in the domain of GAT Ratios, Percentages, Work & Motion Problems. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(6)$.



A 32

B 28 ✓

C 16

D 26

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Ratios, Percentages, Work & Motion Problems):

1. Complete the square: $x^2 - 8x + 28 = (x-4)^2 - 16 + 28 = (x-4)^2 + 12$, giving minimum value $k = 12$. **[M1]**
2. Evaluate at $x = 6$: $f_1(6) = (2)^2 + 12 = 16$. **[M1]**
3. Sum the two quantities: $k + f_1(6) = 12 + 16 = 28$. **[A1]**

- **Final Verified Answer:** 28.

2 MCQ

A mathematical model in GAT Ratios, Percentages, Work & Motion Problems is defined by the cubic function $y_2(x) = 3x^3 - 4x^2 + 15x - 9$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 96

B 57

C 72 ✓

D 34

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Ratios, Percentages, Work & Motion Problems):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 9x^2 - 8x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 9(9) - 8(3) + 15 = 72$. **[M1][A1]**

- **Final Verified Answer:** 72.

3 MCQ

In GAT Ratios, Percentages, Work & Motion Problems, two displacement vectors are given by $\mathbf{u}_3 = \langle 4, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A -20

B 80

C 172

D 60 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Ratios, Percentages, Work & Motion Problems):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (4)(9) + (-8)(7) = 36 + (-56) = -20$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 4^2 + (-8)^2 = 80$. **[M1]**
3. Combine: $-20 + 80 = 60$. **[A1]**

- **Final Verified Answer:** 60.

4 MCQ

The linear growth profile shown in the diagram for GAT Ratios, Percentages, Work & Motion Problems follows an arithmetic progression with first term $a_1 = 5$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 705 ✓

B 89

C 795

D 1335

WORKED SOLUTION & VERIFICATION

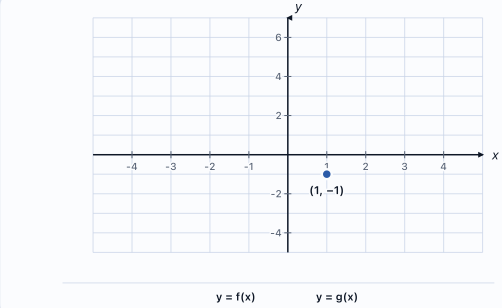
Mark Scheme & Examiner Solution (GAT Ratios, Percentages, Work & Motion Problems):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 5, d = 6$: $S_{15} = \frac{15}{2}[10 + 14 \times 6] = \frac{15}{2} \times 94 = 705$. **[M1][A1]**

- **Final Verified Answer:** 705.

5 MCQ

In GAT Ratios, Percentages, Work & Motion Problems, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 16x)\,dx$.



A 104

B 80 ✓

C 96

D 41

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Ratios, Percentages, Work & Motion Problems):

1. Integrate term by term: $\int (18x^2 + 16x)\,dx = [6x^3 + 8x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 8(2)^2) - 0 = 48 + 32 = 80$. **[M1][A1]**

- **Final Verified Answer:** 80.

6 MCQ

In GAT Ratios, Percentages, Work & Motion Problems, consider the quadratic function $f_6(x) = x^2 - 10x + 109$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(12)$.

A 222

B 133

C 217 ✓

D 210

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Ratios, Percentages, Work & Motion Problems):

1. Complete the square: $x^2 - 10x + 109 = (x-5)^2 - 25 + 109 = (x-5)^2 + 84$, giving minimum value $k = 84$. **[M1]**
2. Evaluate at $x = 12$: $f_6(12) = (7)^2 + 84 = 133$. **[M1]**
3. Sum the two quantities: $k + f_6(12) = 84 + 133 = 217$. **[A1]**

- **Final Verified Answer:** 217.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 3x^3 - 5x^2 + 33x - 16$ studied in GAT Ratios, Percentages, Work & Motion Problems. Find the instantaneous gradient $\frac{dy}{dx}$ of the tangent to the curve at $x = 8$.

A 609	B 496
C 95	D 529 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Ratios, Percentages, Work & Motion Problems):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 9x^2 - 10x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{dy}{dx}\right|_{x=8} = 9(64) - 10(8) + 33 = 529$. **[M1][A1]**

- **Final Verified Answer:** 529.

8 MCQ

In GAT Ratios, Percentages, Work & Motion Problems, two displacement vectors are given by $\mathbf{u}_8 = \langle 5, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 79 ✓	B 5
C 74	D 149

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Ratios, Percentages, Work & Motion Problems):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (5)(8) + (-7)(5) = 40 + (-35) = 5$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 5^2 + (-7)^2 = 74$. **[M1]**
3. Combine: $5 + 74 = 79$. **[A1]**

- **Final Verified Answer:** 79.

9 MCQ

A driver travels a 60 km route from Riyadh to a research station at 30 km/h and returns along the exact same route at 60 km/h. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
60	30

A 13	B 25
C 45 km/h	D 40 km/h ✓

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(30)(60)}{30+60} = 40$ km/h.

10 MCQ

A driver travels a $120\ \text{km}$ route from Riyadh to a research station at $40\ \text{km/h}$ and returns along the exact same route at $60\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
120	40

- **A** $48\ \text{km/h}$ ✓
- **B** $50\ \text{km/h}$
- **C** $53\ \text{km/h}$
- **D** 14

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(40)(60)}{40+60} = 48\ \text{km/h}$.

11 MCQ

A driver travels a $180\ \text{km}$ route from Riyadh to a research station at $60\ \text{km/h}$ and returns along the exact same route at $90\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
180	60

- **A** $70\ \text{km/h}$
- **B** $72\ \text{km/h}$ ✓
- **C** $75\ \text{km/h}$
- **D** $77\ \text{km/h}$

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(60)(90)}{60+90} = 72\ \text{km/h}$.

12 MCQ

A driver travels a $240\ \text{km}$ route from Riyadh to a research station at $60\ \text{km/h}$ and returns along the exact same route at $120\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
240	60

- **A** $85\ \text{km/h}$
- **B** $80\ \text{km/h}$ ✓
- **C** $70\ \text{km/h}$
- **D** $90\ \text{km/h}$

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(60)(120)}{60+120} = 80\ \text{km/h}$.

13 MCQ

A driver travels a $300\ \text{km}$ route from Riyadh to a research station at $20\ \text{km/h}$ and returns along the exact same route at $30\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
300	20

- **A** $24\ \text{km/h}$ ✓
- **B** $29\ \text{km/h}$
- **C** $25\ \text{km/h}$
- **D** $30\ \text{km/h}$

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(20)(30)}{20+30} = 24\ \text{km/h}$.

14 MCQ

A driver travels a $360\ \text{km}$ route from Riyadh to a research station at $50\ \text{km/h}$ and returns along the exact same route at $75\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
360	50

- **A** $60\ \text{km/h}$ ✓
- **B** 14
- **C** $62.5\ \text{km/h}$
- **D** $65\ \text{km/h}$

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(50)(75)}{50+75} = 60\ \text{km/h}$.

15 MCQ

A driver travels a $420\ \text{km}$ route from Riyadh to a research station at $80\ \text{km/h}$ and returns along the exact same route at $120\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
420	80

- **A** $96\ \text{km/h}$ ✓
- **B** $101\ \text{km/h}$
- **C** $90\ \text{km/h}$
- **D** $100\ \text{km/h}$

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(80)(120)}{80+120} = 96\ \text{km/h}$.

16 MCQ

A driver travels a $480\ \text{km}$ route from Riyadh to a research station at $45\ \text{km/h}$ and returns along the exact same route at $90\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
480	45

A $55\ \text{km/h}$

B $60\ \text{km/h}$ ✓

C $67.5\ \text{km/h}$

D $65\ \text{km/h}$

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(45)(90)}{45+90} = 60\ \text{km/h}$.

17 MCQ

A driver travels a $540\ \text{km}$ route from Riyadh to a research station at $60\ \text{km/h}$ and returns along the exact same route at $120\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
540	60

A $70\ \text{km/h}$

B $90\ \text{km/h}$

C $80\ \text{km/h}$ ✓

D $85\ \text{km/h}$

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(60)(120)}{60+120} = 80\ \text{km/h}$.

18 MCQ

A driver travels a $600\ \text{km}$ route from Riyadh to a research station at $80\ \text{km/h}$ and returns along the exact same route at $120\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
600	80

A $90\ \text{km/h}$

B $101\ \text{km/h}$

C $100\ \text{km/h}$

D $96\ \text{km/h}$ ✓

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(80)(120)}{80+120} = 96\ \text{km/h}$.

19 MCQ

A driver travels a $660\ \text{km}$ route from Riyadh to a research station at $120\ \text{km/h}$ and returns along the exact same route at $180\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
660	120

A $144\ \text{km/h}$ ✓

B $149\ \text{km/h}$

C $150\ \text{km/h}$

D $130\ \text{km/h}$

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(120)(180)}{120+180} = 144\ \text{km/h}$.

20 MCQ

A driver travels a $720\ \text{km}$ route from Riyadh to a research station at $120\ \text{km/h}$ and returns along the exact same route at $240\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
720	120

A $180\ \text{km/h}$

B $165\ \text{km/h}$

C $160\ \text{km/h}$ ✓

D $130\ \text{km/h}$

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(120)(240)}{120+240} = 160\ \text{km/h}$.

21 MCQ

A driver travels a $780\ \text{km}$ route from Riyadh to a research station at $40\ \text{km/h}$ and returns along the exact same route at $60\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
780	40

A $53\ \text{km/h}$

B 14

C $48\ \text{km/h}$ ✓

D $50\ \text{km/h}$

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(40)(60)}{40+60} = 48\ \text{km/h}$.

22 MCQ

A driver travels a $840\ \text{km}$ route from Riyadh to a research station at $100\ \text{km/h}$ and returns along the exact same route at $150\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
840	100

A $125\ \text{km/h}$

B $120\ \text{km/h}$ ✓

C 14

D $110\ \text{km/h}$

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(100)(150)}{100+150} = 120\ \text{km/h}$.

23 MCQ

A driver travels a $900\ \text{km}$ route from Riyadh to a research station at $160\ \text{km/h}$ and returns along the exact same route at $240\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
900	160

A $197\ \text{km/h}$

B $200\ \text{km/h}$

C $170\ \text{km/h}$

D $192\ \text{km/h}$ ✓

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(160)(240)}{160+240} = 192\ \text{km/h}$.

24 MCQ

A driver travels a $960\ \text{km}$ route from Riyadh to a research station at $90\ \text{km/h}$ and returns along the exact same route at $180\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
960	90

A $120\ \text{km/h}$ ✓

B $100\ \text{km/h}$

C $135\ \text{km/h}$

D $125\ \text{km/h}$

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(90)(180)}{90+180} = 120\ \text{km/h}$.

25 MCQ

A driver travels a $1020\ \text{km}$ route from Riyadh to a research station at $90\ \text{km/h}$ and returns along the exact same route at $180\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
1020	90

A $100\ \text{km/h}$

B $135\ \text{km/h}$

C $120\ \text{km/h}$ ✓

D $125\ \text{km/h}$

WORKED SOLUTION & VERIFICATION

Round-trip average speed over equal distances is the harmonic mean $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(90)(180)}{90+180} = 120\ \text{km/h}$.